

Cyanotype animation

Cookbook

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Chapter 1

Animation stage and digital preparation

Requirements

Device and programme for digital animation

Mockup and light values

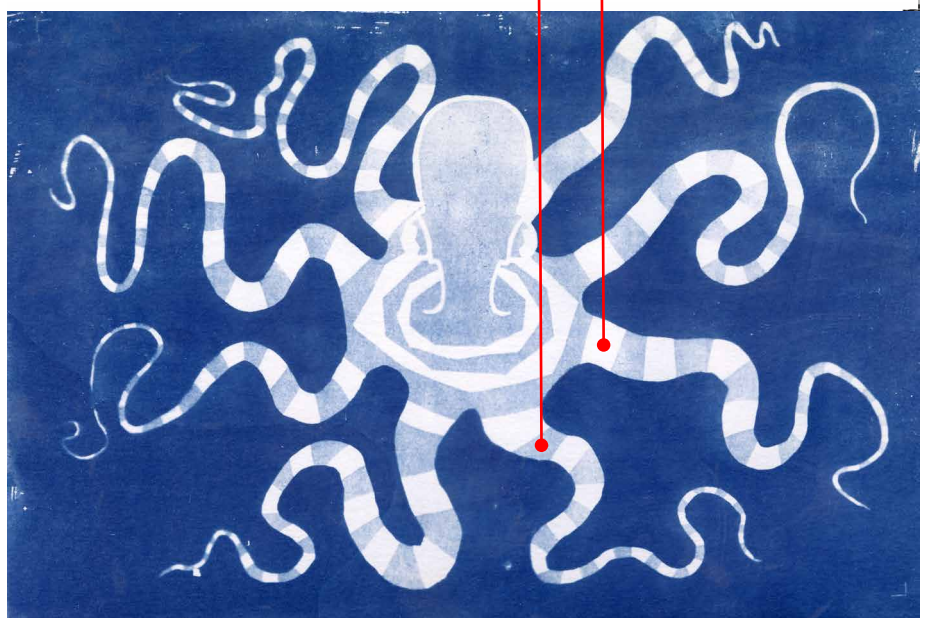
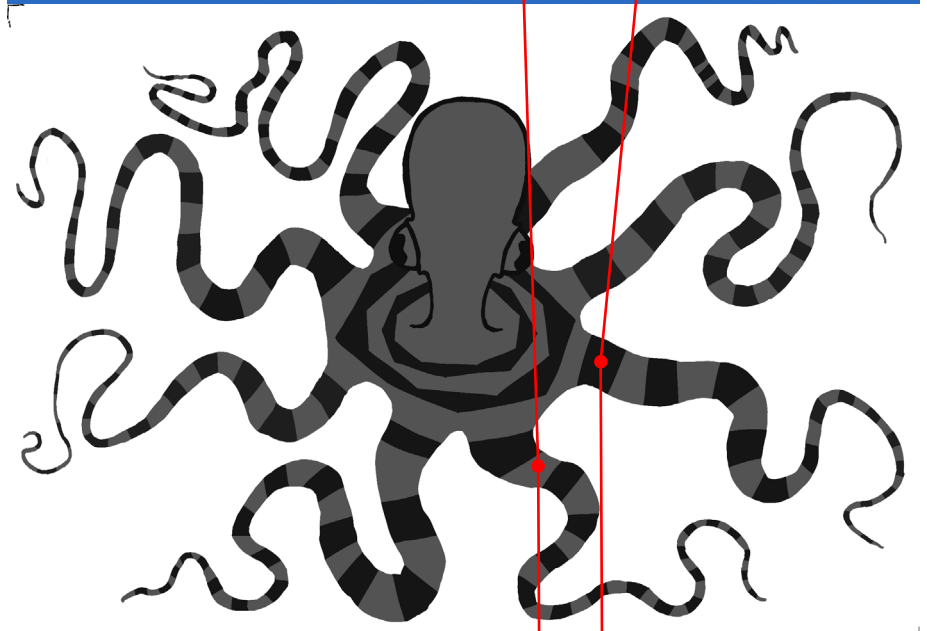
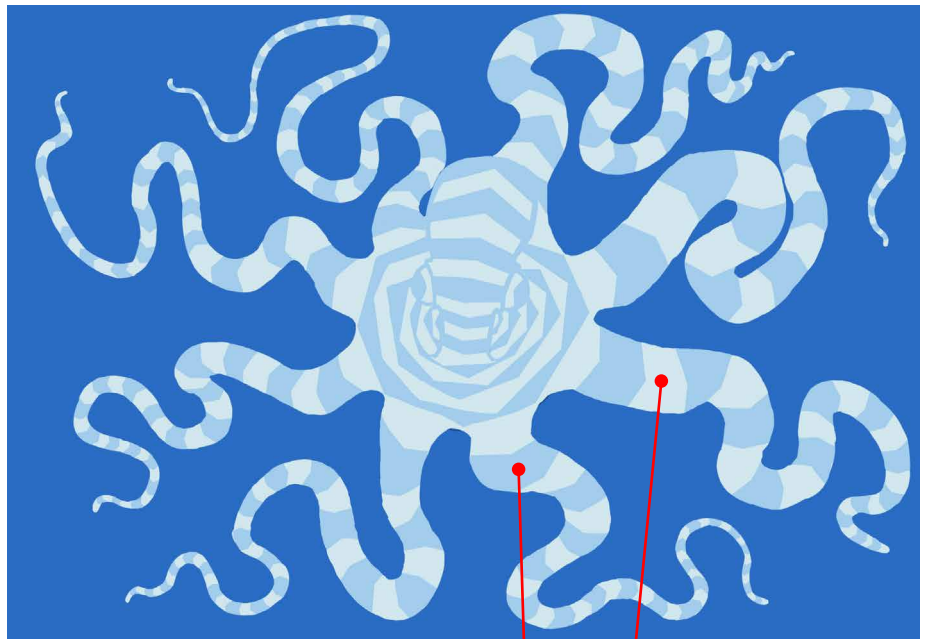
Before you start your digital animation, you'll want to make a digital version of how you roughly want the end product to look, keeping in mind the aesthetic qualities of cyanotype.

After this when making the actual animation, you have to keep in mind that the places where you had drawn lighter values in the mockup, you'll have to draw them darker. After making a mockup, you'll want to start animating. When animating you need to keep in mind that the lighter you want something to be in the final product (like in the mockup), the darker you'll have to make it in the digital animation and vice versa.

Light values in mockup/end product = dark values.

This project can also be done with other light sensitive materials like anthotype or photopaper. For some of these the light value exchange doesn't apply, though for this book we will assume the principles that apply to cyanotype.

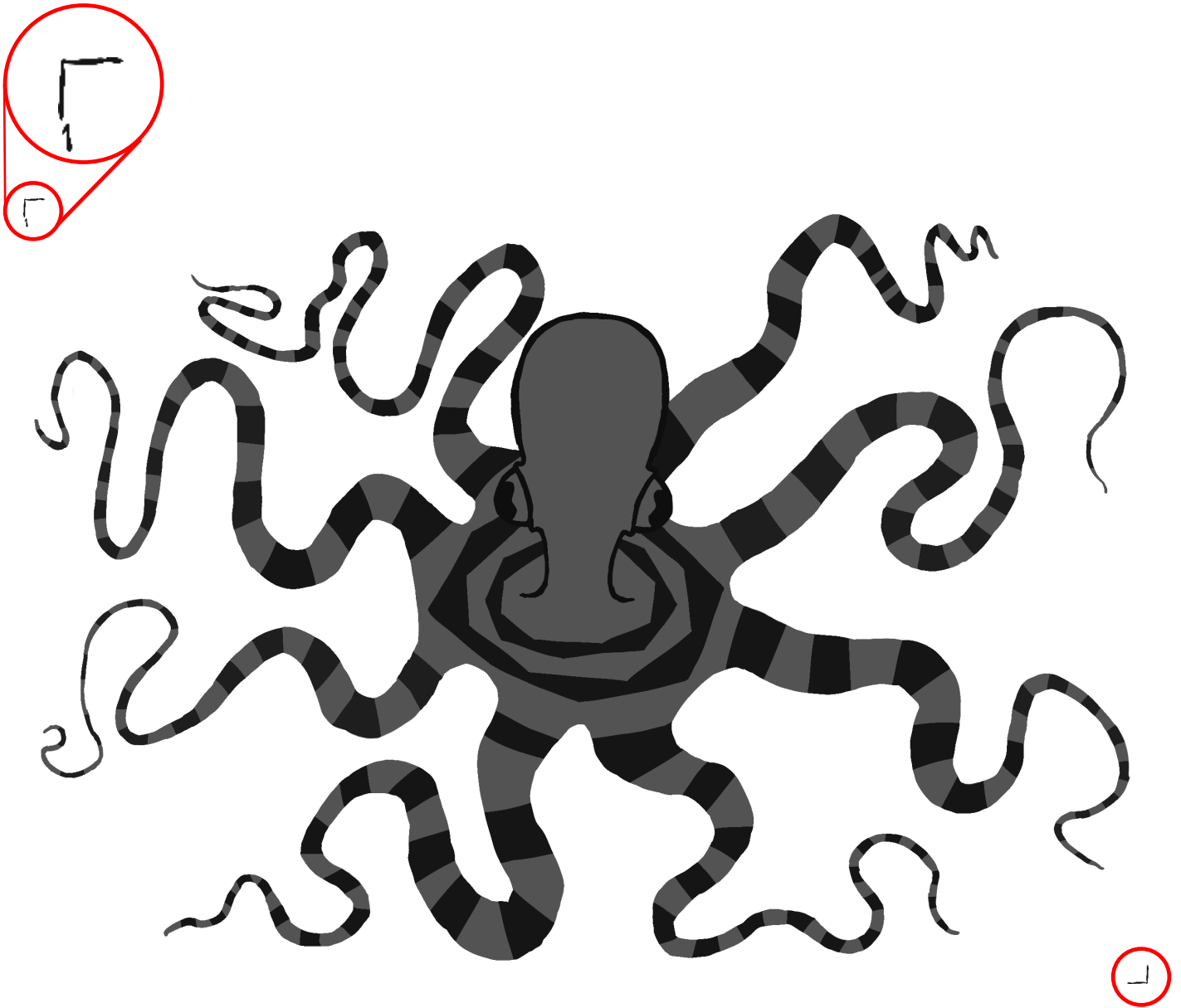
Tip: try to make an animation with a smaller number of frames, smaller loops work perfectly for this.



Frame

Add a frame into the file in each frame where you want the borders of the final animation to be. It's also helpful to add a counter into the file on every frame so it's easy to spot which frame you're working with each step of the way.

You can do it like shown or make an elaborate/larger, more visible border.



Chapter 2

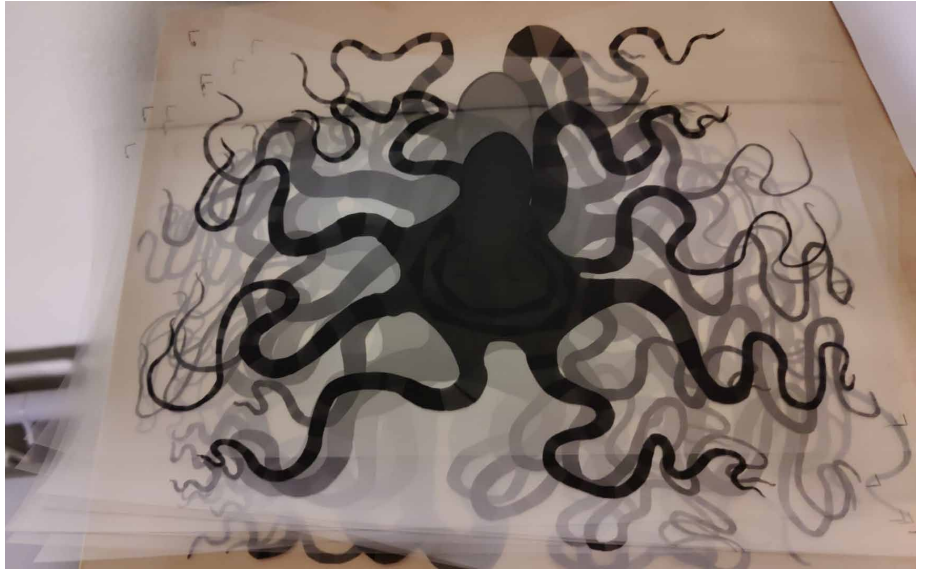
Turning the frames into cyanotype

Requirements

Amount of transparent papers equal to amount of frames of animation you have, UV bed or sunny day, printer access, regular paper, cyanotype medium, cyanotype solution, one or multiple trays, water.

Transparent sheets

Now that you've prepared all your frames of animation digitally you need to export them all separately as images. After that, you print them all out separately on transparent sheets. Before you print the pictures on the sheets however, you first have to do one or multiple test prints by printing a frame on regular paper to make sure they will be printed out correctly.



Cyanotype preparation

Prepare your cyanotype (or other light sensitive material) on your chosen medium. This method is most easily done with larger sheets of watercolor paper, although this could also work on other mediums such as white fabrics.

Apply the cyanotype solution to the paper (or otherwise) in a dark area, it's important to make a couple more than you need. The brushes you're using to apply the solution with shouldn't have any metal as it would react with the chemicals.

Make your own cyanotype solution

You can make your own cyanotype fluid or buy it premade. Here is a quick instruction on how to make it yourself

Materials

25 grams of ammonium iron(III)-citrate
10 grams of potassium ferrocyanide
200 ml of demineralized water (purified water)

Method

Measure all ingredients separately.
Mix powders separately with 100 ml of demineralized water each.
Mix the two solutions right before use.

If you have any left, store it (up to 2 weeks roughly) in a light-tight bottle.

The exposure process

Prepare a tub/tray in a dark area with water.

To expose a frame, lay the transparent sheet with your illustrated frame under either the UV bank or direct sunlight. Lay another piece of glass on top to get a clear image.

Before you expose all of your frames, try exposing a couple one by one to find the right exposure time for your project. This is what the extra papers are for.

The longer you expose something, the darker everything will get, so darker details will show up more clearly and lighter ones less.

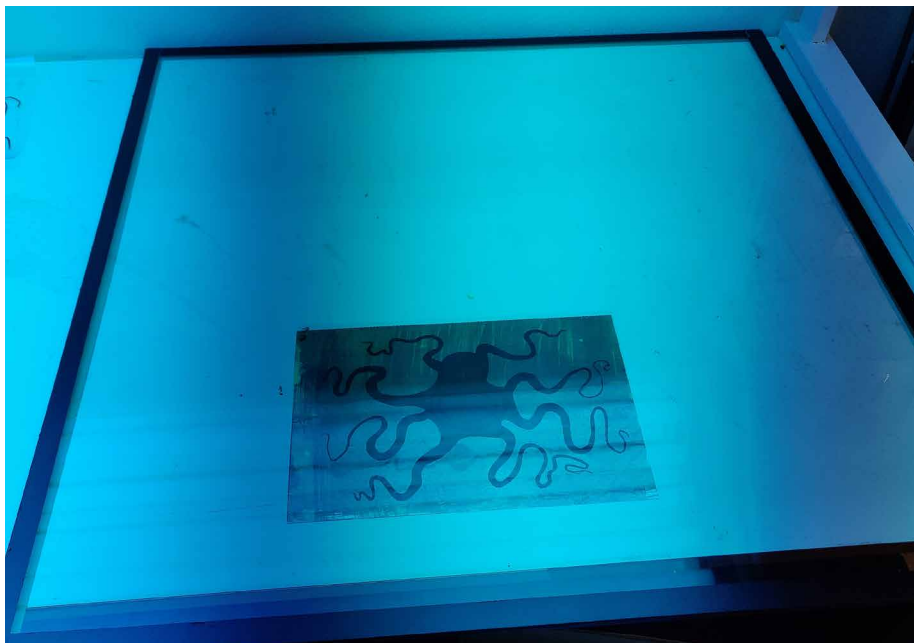
After exposing a frame, quickly bring it into the dark area where you have a tub of water ready to wash away the residual solution on the paper, the image(s) will slowly start showing up.

After you get a feel for how to expose the frames for the amount of time that yields a positive result, you can expose the rest of the frames. You can try exposing a couple at once when you feel confident enough.

Change out the water in the tray if it becomes too saturated with residue.

If you still have any extra's left you can redo any frames you might've messed up or aren't satisfied with.

You can also add another bassin wherein peroxide or vinegar is added into the water. These will influence the final images look, for instance darkening the end result.



Chapter 3

Putting the animation together

Requirements

Scanning device or camera,
animation or video editing software

Digitize

Scan the results of your cyanotype process with either a scanner or a photocopier. If you don't have access to either of these you can try to find a place where you can have good, even lighting and photograph your cyanotypes head on.

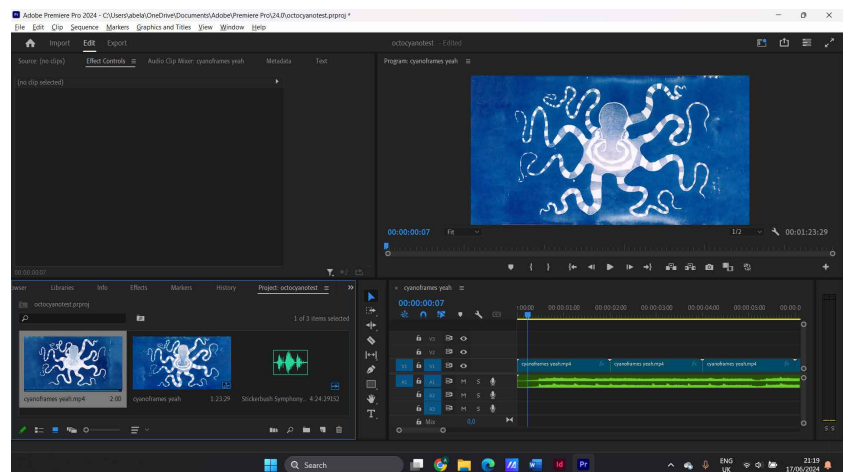
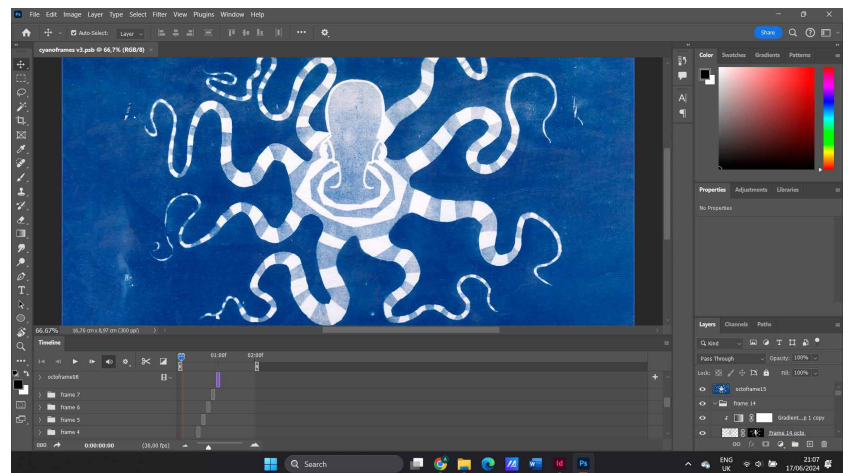
Compiling

Bring all of the digitized files into your editing programme of choice. Once you've imported all your files and put the frames into the right order, align them by the borders you put in at the start and exposed onto the paper.

You start by making only the first frame visible and putting it in the position you want. Next, bring in the next frame or make it viewable and bring down it's opacity. Use the transform tools to align the border markings to bring all the images into the right position. Once they've been aligned bring the opacity back up and repeat for every frame until you've arranged all of them.

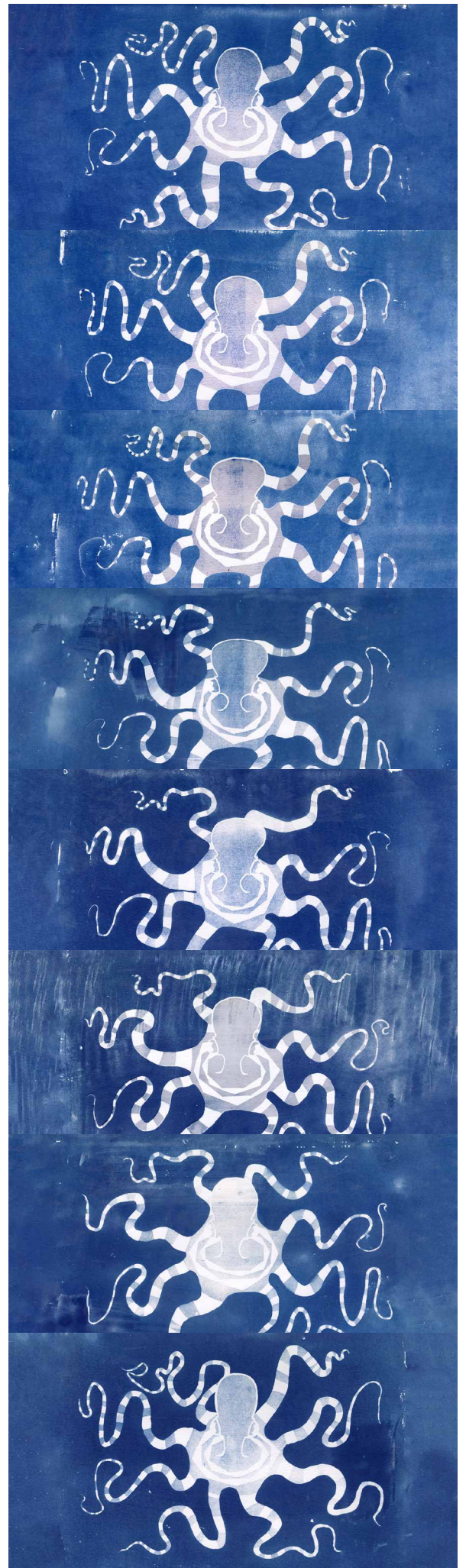
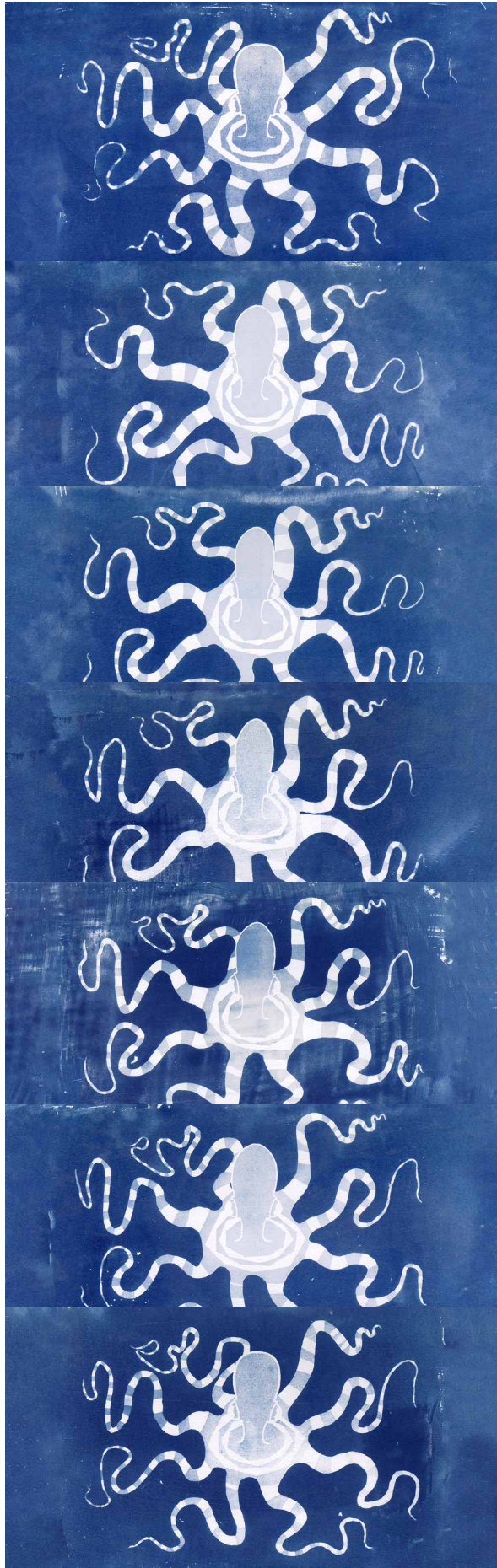
Finalizing

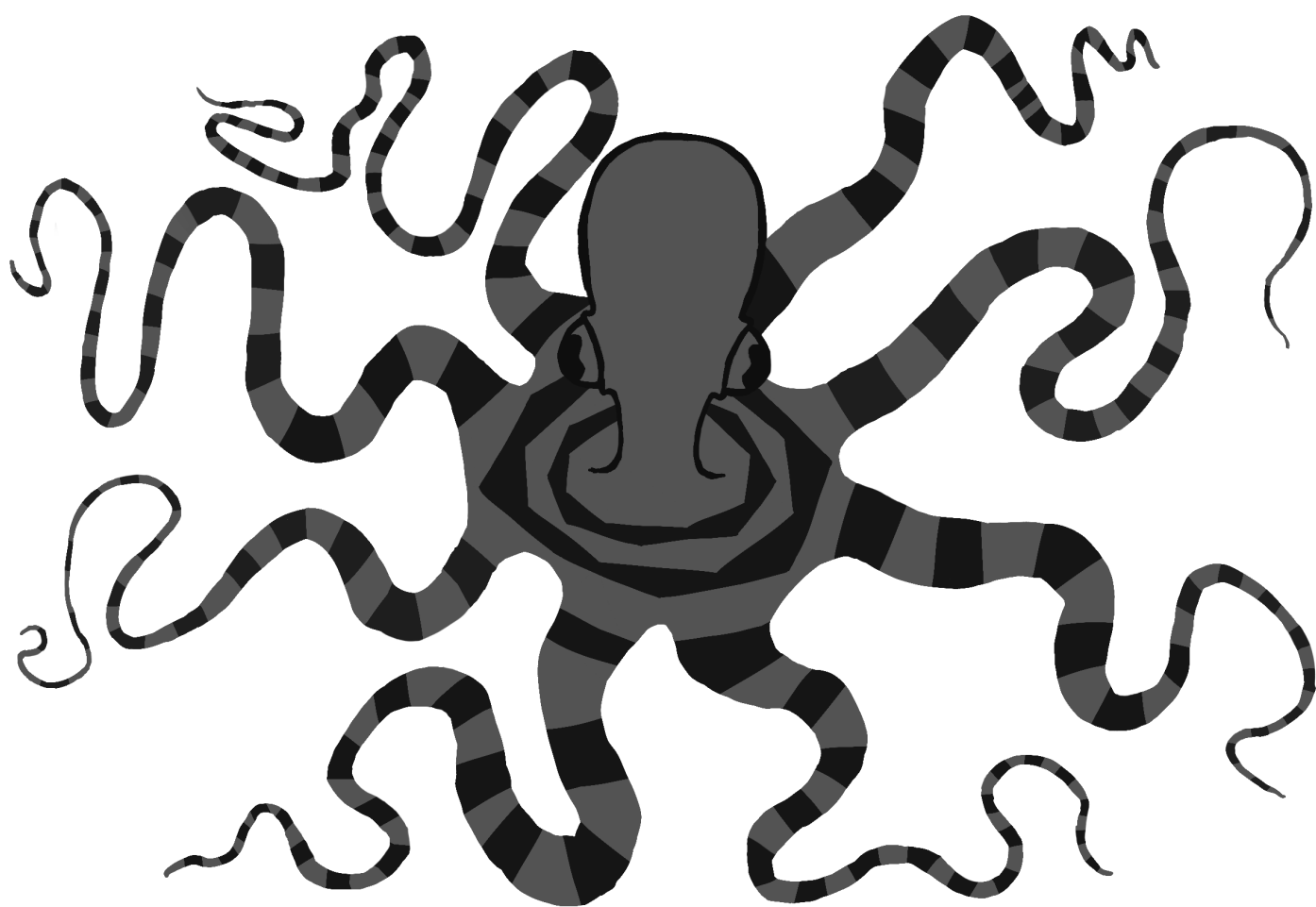
Now you can do any last clean-ups like removing all the borders digitally or expanding the frame, you can now render out the final animation. After rendering the video file, it will probably be very short, you can loop the animation a number of times in your video software. Add any music or sound effects you wish. When you're completely done, render out the final video and you'll have the final result.



In this qr code you'll see my
example of an animation in
cyanotype.

My 15 frames





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Photography otherwise